

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	19 July 2026
Team ID	XXXX XXXX XXXX
Project Name	Ucab – Smart Cab Booking Application
Maximum Marks	4 Marks

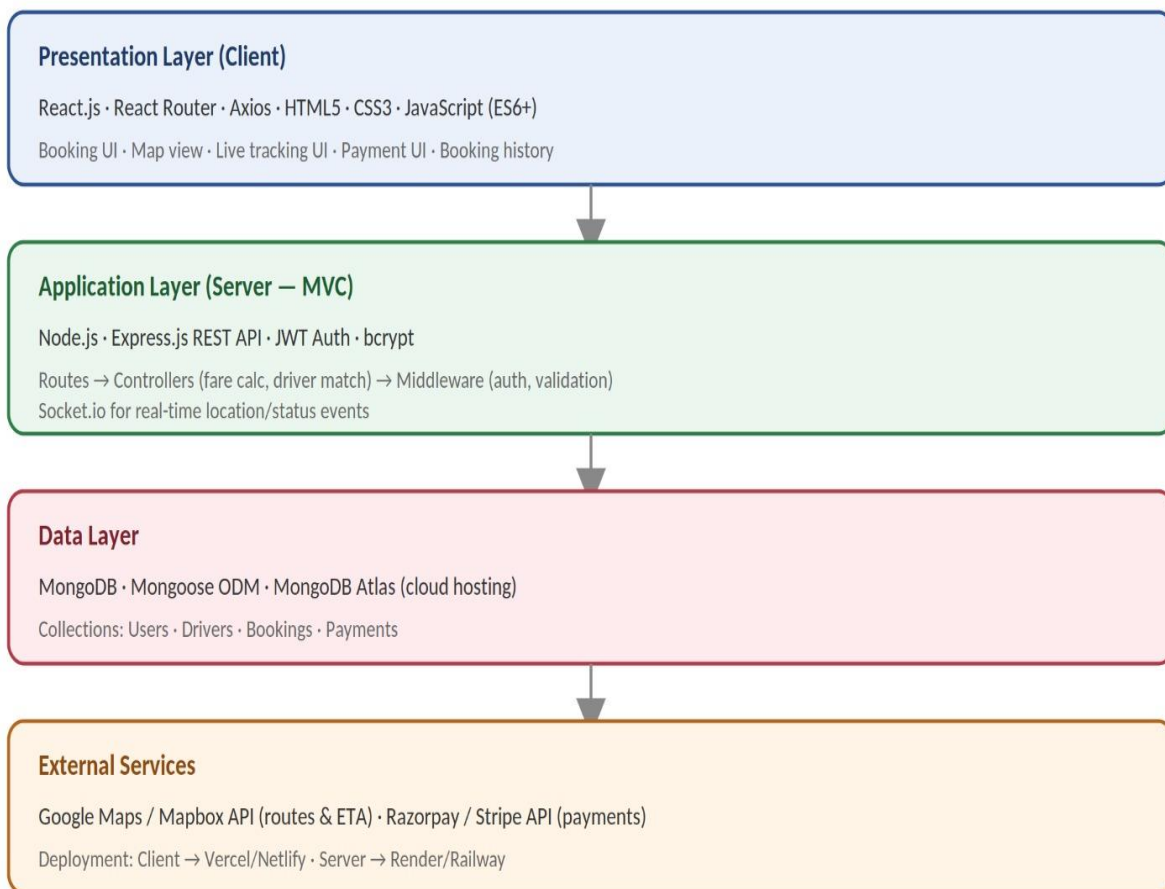
Table 1: Components & Technologies

S.No.	Component	Description	Technology
1	User Interface	Web UI for booking, tracking & payments	HTML, CSS, JavaScript, React.js
2	Application Logic - Client	Client-side state, routing, map interactions	React.js, React Router, Axios
3	Application Logic - Server	REST APIs, business rules, fare calculation	Node.js, Express.js
4	Authentication	User/driver login & session management	JWT, bcrypt
5	Database	Stores users, drivers, bookings, payments	MongoDB (Mongoose ODM)
6	Cloud Database	Managed database hosting	MongoDB Atlas
7	Real-time Updates	Live cab location & trip status	Socket.io / WebSockets
8	External API	Maps, geocoding & route/ETA calculation	Google Maps / Mapbox API
9	Payments	Automated payment processing	Razorpay / Stripe API
10	Infrastructure (Server/Cloud)	Application deployment	Client: Vercel/Netlify; Server: Render/Railway

Table 2: Application Characteristics

S.No.	Characteristics	Description / Technology
1	Open-Source Frameworks	React.js, Express.js, Node.js, Mongoose
2	Security Implementations	JWT auth, bcrypt password hashing, input validation, HTTPS
3	Scalable Architecture	Stateless REST API layer, MVC pattern on backend
4	Availability	Cloud-hosted DB with automated backups (MongoDB Atlas)
5	Performance	Indexed MongoDB queries, lazy-loaded React components

Technology Stack — Ucab



Requests flow top → down through the stack; responses and real-time events flow back up.

